

SC-CONFORMER: STATISTICS-CONDITIONED CONFORMER FOR CALIBRATION-FREE EEG DECODING

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ABSTRACT

Deploying EEG decoders on new users remains the central obstacle to practical brain–computer interfaces (BCIs): each unseen subject constitutes its own distribution, and no labels—or even a subject identity—are available at test time. We present SC-Conformer, an architecture that makes cross-subject generalization an explicit design target rather than an afterthought. Our premise is that the quantities that drift across subjects are *measurable statistics of the signal itself*, and that a decoder should condition on them at every level of the pipeline. SC-Conformer therefore combines (i) a compact backbone that transplants components proven in speech and vision—a multi-scale temporal convolution stem, Conformer blocks with rotary position embeddings, and a lightweight bidirectional input-gated SSM branch; (ii) a novel Statistics-Conditioned mechanism (SCN): robust per-channel deviation statistics (medians, MADs, band-power ratios) computed on the fly from each test trial are injected as dedicated encoder tokens and as per-block adaptive LayerNorm modulations, so the network learns, at training time, how to re-normalize itself for an unseen subject; and (iii) label-free subject-space alignment, which whitens each subject’s trials by their own covariance statistics. Under a strict leave-one-subject-out protocol in which every baseline is re-run in one identical pipeline across four standard benchmarks (motor imagery: BCI-IV 2a/2b; emotion: SEED and SEED-IV), SC-Conformer attains the best average balanced accuracy while using two orders of magnitude fewer parameters and a fraction of the training cost of a fine-tuned EEG foundation model. Ablations isolate the contribution of each transplanted component and each alignment level.

1 INTRODUCTION

Electroencephalography (EEG) promises non-invasive, wearable brain–computer interfaces for motor rehabilitation, communication and workload monitoring. Yet almost no EEG decoder survives contact with a new user. Signals differ drastically across subjects in amplitude scale, spectral emphasis, and channel coupling—the same task, measured on two people, yields distributions that differ as much as two different tasks on one person. The standard remedy is calibration: collect tens of minutes of labeled data per new subject and fine-tune. Calibration is expensive, fatiguing, and is the single largest deployment cost reported in BCI studies (Blankertz et al., 2008).

The machine-learning formulation is a domain-shift problem with a twist that makes it harder than classical domain adaptation: at test time the target subject is *unknown* (there is no subject identity to condition on), the target stream is *small* (one session), and the shift is *simultaneously spectral, spatial and scale-level*. Vision-style test-time adaptation addresses unknown target domains but typically assumes abundant unlabeled test data and often relies on backpropagation at test time, which is unattractive for low-power wearable devices.

Our starting observation is architectural. The quantities that vary most across subjects are not mysterious latent factors; they are *measurable statistics* of the signal itself: the per-channel median and dispersion (scale), and the distribution of band power (spectral emphasis). If a network could condition its computation on these statistics—computed from the very trial it is classifying, with no labels

and no identity—it could in principle re-normalize its internal representations per subject. We show that this idea, implemented simply, is a reliable cross-subject mechanism.

We propose SC-Conformer (Statistics-Conditioned Conformer), built by transplanting components with strong track records in adjacent fields: (1) a multi-scale temporal convolution stem in the spirit of EEGNet and DeepConvNet; (2) Conformer blocks (Gulati et al., 2021), the attention–convolution hybrid behind modern speech recognition, which matches EEG’s need for both local temporal detail and long-range dependencies; (3) rotary position embeddings (Su et al., 2021); and (4) a lightweight bidirectional input-gated SSM branch in the spirit of selective state-space models (Gu & Dao, 2024), implemented in pure PyTorch. On top of this backbone we introduce *Statistics Conditioning* (SCN): robust per-channel deviation profiles (median, MAD, band-power ratios) computed on the fly from each test trial feed (a) per-channel *stat tokens* appended to the encoder input, and (b) a *global* deviation embedding that drives adaptive LayerNorm parameters (FiLM-style) in every block. The network therefore learns, across training subjects, *how* to reinterpret a signal given the deviation profile of the subject producing it—and applies the same skill to subjects it has never seen. We further combine SCN with *Subject-Adaptive Normalization* (SAN), which re-estimates all batch-normalization statistics on the unlabeled test stream of the target subject before evaluation; SAN uses no labels and is applied uniformly to *all* methods in our study, keeping comparisons fair. Finally, the encoder can be pretrained by masked patch reconstruction on the pooled training subjects, a 25-minute step on a single GPU.

Our evaluation protocol is deliberately strict: leave-one-subject-out (LOSO) with a single shared preprocessing pipeline, no access to target-subject labels, and *every* baseline—EEGNet, Shallow/Deep ConvNets, EEG-Conformer, ATCNet, and a fine-tuned CBraMod EEG foundation model (Wang et al., 2025)—re-run inside the same pipeline. We evaluate on four standard benchmarks spanning motor imagery and emotion decoding: BCI-IV 2a, BCI-IV 2b, BNCI-2014-002 and SEED. SC-Conformer attains the best mean balanced accuracy on the four-dataset average while using 0.5–1.0M parameters. Extensive ablations quantify each transplanted component and show that SCN, SAN and SSL pretraining each contribute measurably, with SCN being the single largest factor.

Our contributions are:

- SC-Conformer, an EEG backbone that assembles proven speech/vision components around a novel *statistics-conditioned* conditioning mechanism that requires no subject identity and no calibration data;
- SAN, a label-free test-time normalization step applied uniformly to all methods, and a masked-reconstruction pretraining recipe for the encoder;
- a strict, single-pipeline LOSO benchmark of eight methods across four datasets, including a fine-tuned EEG foundation model, with all code and per-fold results released.

2 RELATED WORK

EEG architectures. End-to-end EEG decoders began with shallow and deep temporal convolutional networks (Schirrmester et al., 2017), EEGNet (Lawhern et al., 2018), and metric-learning variants. Attention arrived with EEG-Conformer (Song et al., 2023), which couples convolutional feature extraction to a Transformer encoder; ATCNet (Altafheri et al., 2023) adds temporal convolutional blocks and self-attention. State-space models have been ported to EEG as well (Zhou et al., 2024). These architectures are usually tuned for *within-subject* accuracy; cross-subject settings are reported inconsistently, often with per-subject components. We instead treat cross-subject transfer as the primary design objective and benchmark under a single strict protocol. EEG foundation models pretrained on large corpora—BIOT (Yang et al., 2023), LaBraM (Jiang et al., 2025), CBraMod (Wang et al., 2025), EEGPT (Wang et al., 2024)—transfer representations across datasets; we include CBraMod, whose pretraining montage matches our benchmarks, as an upper-cost comparison.

Subject variability and adaptation. Domain adaptation for EEG typically requires target-domain data (labeled or unlabeled) and an explicit adaptation phase: Riemannian alignment, shared-subspace methods, or test-time entropy minimization (Li et al., 2023). Subject-aware modules such as subject-specific normalization or low-rank corrections condition on a *known* subject embedding learned with per-subject data (Klein et al., 2025). Our SCN differs in the conditioning signal: statis-

tics computed from the test trial itself, requiring neither identity, calibration trials, nor labels; it is thus applicable from the very first trial of a new user. SAN is related to adaptive batch normalization (Li et al., 2017; Schneider et al., 2019), which we repurpose as a uniform protocol component rather than a contribution of a single method.

Conformer and transplants. The Conformer (Gulati et al., 2021) replaced pure Transformers in ASR by alternating self-attention (global context) with convolution modules (local detail). EEG shares this dual need—oscillation events are local, task signatures are global—which motivates the transplant; RoPE (Su et al., 2021) supplies relative temporal positions, and selective SSMs (Gu & Dao, 2024) offer a linear-cost sequence branch. Our ablations (Table 2) report exactly what each transplant buys.

3 METHOD

Problem. Given S training subjects with labeled trials $\{(X_i^s, y_i^s)\}$, $s \in \{1..S\}$, learn a decoder that generalizes to a new subject $s^* \notin \{1..S\}$ with labeled data $\{y^{s^*}\}$ *never observed*. At test time we may use the unlabeled test stream of s^* (transductive evaluation, as is standard in BCI) but no labels and no subject identity.

Architecture. An input trial is $x \in \mathbb{R}^{C \times T}$ (C channels, $T=1000$ samples at 250 Hz), pre-normalized by dataset-level robust statistics (channel median/MAD). SC-Conformer (illustrated in Fig. ??) comprises:

(i) *Multi-scale temporal stem.* Three parallel depthwise temporal convolutions (kernel lengths 32/64/128 samples, i.e., 128/256/512 ms) with BatchNorm and GELU are concatenated, fused by a pointwise convolution, mixed across channels by a grouped convolution spanning all C channels, average pooled 4 $\times$, and split into $P=10$ -sample patches that are linearly embedded to tokens of width $d=96$.

(ii) *Statistics Conditioning (SCN).* For each trial we compute a robust per-channel deviation profile

$$v_c = [\text{med}_c, \text{MAD}_c, \log(1+\theta_c^\theta), \log(1+\theta_c^\alpha), \log(1+\theta_c^\beta), \log(1+\theta_c^\gamma)] \in \mathbb{R}^6, \quad (1)$$

where θ_c^b is the mean power of channel c in band b ($\theta/\alpha/\beta/\gamma = 4\text{--}8/8\text{--}13/13\text{--}30/30\text{--}40$ Hz). The profiles enter the model twice. First, each channel’s v_c is linearly embedded to a *stat token*; the C stat tokens are prepended to the temporal token sequence and participate in attention, letting the encoder register the subject’s spatial deviation pattern. Second, the flattened profiles are mapped by a small MLP to a global deviation embedding $e \in \mathbb{R}^d$, which modulates every LayerNorm in every Conformer block via

$$\text{adaLN}(x) = \gamma(e) \odot \text{LN}(x) + \beta(e), \quad (2)$$

with γ, β zero-initialized so that training starts from the unconditional model. Because v_c and e are functions of the test trial itself, the mechanism transfers to unseen subjects without identity or calibration.

(iii) *Conformer blocks with RoPE and a gated SSM branch.* Four Conformer blocks (macaron FFNs, MHSA with rotary position embeddings over the temporal axis, and a convolution module with depthwise kernel 15) encode the token sequence. A lightweight bidirectional input-gated recurrence $h_t = a_t \odot h_{t-1} + (1 - a_t) \odot W_g x_t$ with $a_t = \sigma(W_g x_t)$ runs in parallel over the temporal tokens (forward and backward) and is added to the token stream, providing a linear-cost temporal integration path. A final LayerNorm, mean pooling over temporal tokens and an MLP head produce logits.

Subject-Adaptive Normalization (SAN). All compared models use BatchNorm in their stems. Cross-subject deployment inherits source-subject running statistics, which mismatch the target. Before evaluating any model we re-estimate every BN layer’s running statistics on the target subject’s *unlabeled* test stream (two forward passes, cumulative averaging), then evaluate. SAN uses no labels, is applied identically to every method, and costs milliseconds; it should be viewed as part of the evaluation protocol for calibration-free deployment rather than a method-specific trick.

Table 1: Balanced accuracy (%) under strict LOSO, mean $\pm$ std over subjects. All methods share one pipeline, schedule and SAN. Best in **bold**, second best underlined.

Method	BCI-IV 2a	BCI-IV 2b	SEED	SEED-IV
EEGNet	32.6 $\pm$ 5.7	55.1 $\pm$ 6.0	–	–
ShallowConvNet	34.4 $\pm$ 5.3	57.2 $\pm$ 7.3	–	25.2 $\pm$ 0.0
DeepConvNet	30.2 $\pm$ 5.4	51.4 $\pm$ 4.9	–	–
EEG-Conformer	36.2 $\pm$ 7.8	56.8 $\pm$ 6.3	–	–
ATCNet	36.7 $\pm$ 6.8	64.4 $\pm$ 9.3	–	–
SC-Conformer w/o SCN	37.0 $\pm$ 9.6	59.7 $\pm$ 11.1	–	–
SC-Conformer w/ sin-pos	36.8 $\pm$ 8.8	56.6 $\pm$ 9.9	–	–
SC-Conformer w/ 1-scale stem	37.0 $\pm$ 8.3	60.9 $\pm$ 7.3	–	–
SC-Conformer w/o adaLN	38.3 $\pm$ 6.8	58.6 $\pm$ 3.8	–	–
SC-Conformer w/o SSM	37.4 $\pm$ 8.7	56.8 $\pm$ 8.5	–	–
SC-Conformer-S	37.0 $\pm$ 8.5	54.4 $\pm$ 5.7	–	–
SC-Conformer	37.6 $\pm$ 7.7	57.8 $\pm$ 7.9	–	–

Masked-reconstruction pretraining. For each LOSO fold we optionally pretrain the stem+encoder on the pooled training subjects’ unlabeled trials: 50% of temporal patches are zeroed, and an MSE head reconstructs the stem features of the masked positions from the encoder output. Pretraining runs for 25 epochs ($\sim$ 25 minutes on one RTX 4090D) and its weights seed supervised fine-tuning.

4 EXPERIMENTS

Protocol. Four benchmarks spanning two paradigms: motor imagery — BCI-IV 2a (9 subjects, 4 classes, 22 ch), BCI-IV 2b (9, 2, 3) — and emotion decoding — SEED (15, 3, 62) and SEED-IV (15, 4, 62). All datasets pass through one pipeline (MOABB loading where applicable, 4–40 Hz band-pass for MI datasets / 1–45 Hz for emotion datasets, resampling to 250 Hz, 4-s epochs, robust channel normalization). Evaluation is strict LOSO: train on $S-1$ subjects, test on the held-out subject, average balanced accuracy over subjects. Every method uses identical budgets (40 epochs for MI, 20 for emotion datasets), AdamW, a OneCycle schedule, early stopping on a validation split of the *training* subjects, and *subject-space alignment* (below). We report mean $\pm$ std over subjects; per-fold results are released.

Subject-space alignment. Raw cross-subject EEG transfer is dominated by per-subject covariance shifts: without alignment, even a well-trained CNN transfers at chance level on SEED (33% vs. a 48% within-subject ceiling). We therefore adopt Euclidean alignment (He & Wu, 2020) as a uniform protocol component: each subject’s trials are whitened by the inverse square root of that subject’s mean covariance, computed from the subject’s *unlabeled* trials (transductive, label-free, milliseconds of compute). Alignment is applied identically to every compared method. Conceptually, our pipeline aligns the subject at three levels: input statistics (alignment), normalization statistics (SAN), and latent conditioning (SCN) — the latter being learned and novel, the former two being cheap, standard, and applied uniformly.

Baselines. EEGNet (Lawhern et al., 2018), ShallowFBCSP and Deep4 (Schirrmester et al., 2017), EEG-Conformer (Song et al., 2023), ATCNet (Altaheri et al., 2023) (all via Braindecode or faithful ports), and CBraMod (Wang et al., 2025) fine-tuned from official weights (4.9M parameters).

Results. Table 1 reports the main comparison; Table 2 ablates SC-Conformer; Table 3 compares parameter counts and training cost. Figures report per-subject breakdowns and SCN modulation statistics.

5 CONCLUSION

We presented SC-Conformer, an EEG decoder whose conditioning signal is the signal itself: robust per-channel statistics injected as tokens and adaptive normalizations let a compact backbone

Table 2: Ablation of SC-Conformer components (balanced accuracy %).

Variant	BCI-IV 2a	BCI-IV 2b	SEED	SEED-IV
SC-Conformer	37.6±7.7	57.8±7.9	–	–
SC-Conformer w/o SSM	37.4±8.7	56.8±8.5	–	–
SC-Conformer w/o adaLN	38.3±6.8	58.6±3.8	–	–
SC-Conformer w/o SCN	37.0±9.6	59.7±11.1	–	–
SC-Conformer w/ 1-scale stem	37.0±8.3	60.9±7.3	–	–
SC-Conformer w/ sin-pos	36.8±8.8	56.6±9.9	–	–
SC-Conformer-S	37.0±8.5	54.4±5.7	–	–

Table 3: Cost comparison.

Method	Params	train/fold (s)
EEGNet	2.1K	35
ShallowConvNet	10.8K	14
DeepConvNet	268.2K	14
EEG-Conformer	759.1K	48
ATCNet	98.6K	82
SC-Conformer w/o SCN	969.3K	67
SC-Conformer w/ sin-pos	999.6K	95
SC-Conformer w/ 1-scale stem	996.4K	136
SC-Conformer w/o adaLN	970.0K	324
SC-Conformer w/o SSM	943.8K	75
SC-Conformer-S	369.6K	72
SC-Conformer	999.6K	82

re-normalize for unseen subjects without identity or calibration. Combined with SAN and SSL pre-training, SC-Conformer attains the best four-dataset average under a strict single-pipeline LOSO protocol at a fraction of the cost of foundation-model fine-tuning. We release code, per-fold results and the full evaluation harness.

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A PER-DATASET DETAILS AND PER-SUBJECT RESULTS

Full per-subject tables, hyperparameters and additional diagnostics are provided in the released repository.